Баранько Андрій Група Б6236

Варіант Б17

а) $y = e^x \cdot 6 \sin x$

$y = e^x \cdot 6 \sin x = 6 \cdot e^x \cdot \sin x$

$y' = 6 \cdot (e^x \cdot \sin x)' = 6 \cdot ((e^x)' \cdot \sin x + e^x (\sin x)')$

$= 6 \cdot (e^x \cdot \sin x + e^x \cos x) = 6 \cdot e^x \cdot (\sin x + \cos x)$

$y'' = 6 \cdot (e^x \cdot (\sin x + \cos x))' =$

$= 6 \cdot [(e^x)' \cdot (\sin x + \cos x) + e^x (\sin x + \cos x)'] =$

$= 6 \cdot [e^x (\sin x + \cos x) + e^x (\cos x - \sin x)] =$

$= 6 \cdot [e^x \cdot \sin x + e^x \cdot \cos x + e^x \cos x - e^x \cdot \sin x] =$

$= 6 \cdot 2 \cdot e^x \cdot \cos x = 12 e^x \cdot \cos x$

Відповідь: $12 e^x \cdot \cos x$.

б) $2xy + \cos y = 7$

I. $(2xy)'_x + (\cos y)'_x = 7'$

$(2xy)'_x = 2y + 2xy'$

$(\cos y)'_x = \cancel{\ldots} - \sin y \cdot y'$

$2y + 2xy' - y' \sin y = 0$

$2xy' - y' \cdot \sin y = -2y$

$y'(\sin y - 2x) = 2y$

$y' = \dfrac{2y}{\sin y - 2x}$

II $(y')'_x = \dfrac{(2y)'_x \cdot (\sin y - 2x) - 2y \cdot (\sin y - 2x)'_x}{(\sin y - 2x)^2} =$

$= \dfrac{2y' \cdot (\sin y - 2x) - 2y \cdot (y' \cos y - 2)}{(\sin y - 2x)^2} =$

$= \dfrac{2y' \sin y - 4xy' - 2yy' \cos y + 4y}{(\sin y - 2x)^2}$

$$y''_x = \frac{y'(2\sin y - 4x - 2y\cos y) + 4y}{(\sin y - 2x)^2} =$$

$$= \frac{\frac{2y}{\sin y - 2x}\cdot(2\sin y - 4x - 2y\cos y) + 4y}{(\sin y - 2x)^2} =$$

$$= \frac{2y(2\sin y - 4x - 2y\cos y) + 4y(\sin y - 2x)}{(\sin y - 2x)(\sin y - 2x)^2} =$$

$$= \frac{4y\sin y - 8xy - 4y^2\cos y + 4y\sin y - 8xy}{(\sin y - 2x)^3} =$$

$$= \frac{8y\cdot\sin y - 4y^2\cos y - 16xy}{(\sin y - 2x)^3}$$

б) $\begin{cases} x = \arcsin 2t \\ y = 4t^2 - 6 \end{cases}$

$$y'_x = \frac{(4t^2-6)'}{(\arcsin 2t)'} = \frac{8t}{\frac{2}{\sqrt{2-(2t)^2}}\cdot 2t'} = 8t\cdot\frac{\sqrt{2-4t^2}}{2} =$$

$$= 4t\cdot\sqrt{2-4t^2}$$

$$y''_x = \frac{(4t\cdot\sqrt{2-4t^2})'}{(\arcsin 2t)'} = \left\{(4t\cdot\sqrt{2-4t^2})' = 4\cdot\sqrt{2-4t^2} +\right.$$

$$\left. + 4t\cdot\frac{2}{2\cdot\sqrt{2-4t^2}}\cdot(-8t) = 4\cdot\sqrt{2-4t^2} - \frac{16t^2}{\sqrt{2-4t^2}}\right\} =$$

$$= \left(4\sqrt{2-4t^2} - \frac{16t^2}{\sqrt{2-4t^2}}\right) : \frac{2}{\sqrt{2-4t^2}} = \left(4\cdot\sqrt{2-4t^2} - \frac{16t^2}{\sqrt{2-4t^2}}\right)\cdot$$

$$\cdot\frac{\sqrt{2-4t^2}}{2} = \frac{4\cdot\sqrt{2-4t^2}}{2} - \frac{\sqrt{2-4t^2}}{2} - \frac{16t^2}{\sqrt{2-4t^2}}\cdot\frac{\sqrt{2-4t^2}}{2}$$

$$= 2\cdot(2-4t^2) - 8t^2 = 2 - 8t^2 - 8t^2 = 2 - 16t^2$$

Ответ: $2 - 8t^2 - 8t^2 = 2 - 16t^2$